

Evaluating LLMs on Taiwanese Proverbs: A Study of Interpretation and Semantic Understanding

Bachelor's Thesis

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Declaration of Originality

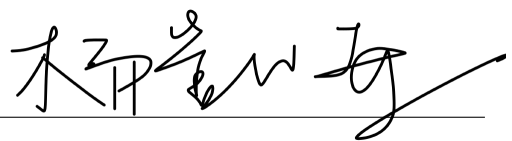
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Abstract

Taiwanese proverbs are an essential part of Taiwan’s language and cultural heritage, often relying on metaphor, cultural knowledge, and contextual interpretation. This study examines how large language models (LLMs) interpret and apply Taiwanese proverbs in English contexts through two complementary evaluation tasks.

Four LLMs were evaluated on two tasks: an Interpretation task, which assessed their ability to explain the meanings of Taiwanese proverbs, and a Context Application task, which examined their ability to select the most appropriate proverb for a given context. The evaluation included 468 Taiwanese proverbs for the Interpretation task and 100 context-based questions for the Context Application task.

In the Interpretation task, Qwen 3.7 achieved the highest mean BERTScore-F1 score (0.9012), followed by GPT-5.6 Luna (0.8947), Gemini 3.6 Flash (0.8859), and DeepSeek (0.8825). The differences between the models were statistically significant. In the Context Application task, Gemini 3.6 Flash achieved the highest accuracy (98%), followed by Qwen 3.7 and DeepSeek (94%), while GPT-5.6 Luna achieved 83%.

These findings indicate that model performance varied across the two evaluation tasks, suggesting that proverb comprehension involves different abilities and should therefore be assessed through complementary tasks. All data, model outputs and prompts are available in this project’s GitHub repository.¹

¹<https://github.com/yangkathyde/bachelor-thesis-taiwanese-proverbs>

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Chapter 1

Introduction

1.1 Background and Research Problem

Figurative language is commonly used in everyday communication, but it can be difficult for language models to understand. Unlike literal expressions, figurative expressions often cannot be understood only from the meanings of individual words. Their interpretation may require linguistic knowledge, contextual information, and knowledge of conventionalized meanings (Shutova, 2011). Proverbs are a particularly important form of figurative language because they commonly express general observations, advice, or social values through culturally conventionalized expressions (Mieder, 2004).

With the development of large language models (LLMs), their ability to understand figurative language has become increasingly important. LLMs have demonstrated strong capabilities across a wide range of natural language processing (NLP) tasks, including language understanding and generation (Zhao et al., 2026). However, strong performance on general language tasks does not necessarily guarantee reliable understanding of figurative language. Recent research suggests that model performance on figurative-language tasks may be influenced by surface-level features rather than by a deep understanding of figurative meaning (Sanchez-Bayona & Agerri, 2025).

Figurative language requires models to infer meanings that may not be directly recoverable from literal wording. For culturally embedded expressions such as proverbs, interpretation may additionally require contextual and cultural knowledge. Previous research has also reported limitations in the ability of multilingual language models to capture language-specific and culturally dependent meanings (Liu et al., 2024).

Taiwanese Southern Min (hereafter *Taigi*) is a variety of Southern Min spoken in Taiwan. Its linguistic features and interpretation of meaning have been examined from various perspectives, including the interaction between lexical meaning, syntactic constructions, and contextual information (Lien, 2003). Taigi proverbs offer a useful domain for studying culturally embedded figurative language. Understanding these expressions may require more than interpreting the literal meanings of individual words, as their intended meanings can depend on conventional interpretations and contextual or cultural knowledge (Mieder, 2004; Shutova, 2011).

Although LLMs have shown strong performance on many general language understanding tasks, their ability to interpret and appropriately apply culturally embedded expressions remains an important area of investigation. In particular, understanding the meaning of a proverb in isolation does not necessarily demonstrate that a model can recognize when and how the proverb should be appropriately used in a particular situation. This difference highlights the importance of evaluating both the way models understand the meanings of proverbs and the way they apply proverbs in different contexts.

1.2 Research Gap

Although recent research has explored LLM capabilities in language understanding, figurative language processing, and multilingual reasoning, several research gaps remain. Previous research has shown that LLMs can perform well on a wide range of language understanding tasks (Zhao et al., 2026). However, studies on figurative language have also identified limitations in the ability of models to interpret figurative meanings, particularly when interpretation requires more than the reliance on surface-level linguistic patterns (Sanchez-Bayona & Agerri, 2025). These challenges are particularly relevant to proverb understanding because the interpretation of a proverb may depend on conventionalized meanings, cultural knowledge, and the context in which it is used (Attia et al., 2026; Mieder, 2004).

First, culturally embedded and low-resource language varieties, such as Taiwanese Taigi, remain underrepresented in NLP research and evaluation. Recent work on constructing LLMs for Taigi and Hakka explicitly identifies the scarcity of linguistic resources as a major challenge (Y.-C. Lai et al., 2024). However, the evaluation of LLMs on cul-

turally embedded Taiwanese proverbs remains relatively underexplored. In particular, further investigation is needed to understand how effectively LLMs can interpret and apply proverbs in a culturally specific and low-resource language context.

Second, the Interpretation task and the Context Application task may involve different but complementary aspects of language understanding. When a model explains a proverb, it shows whether the model understands its intended meaning. When it chooses a proverb for a specific situation, it shows whether it can apply that meaning appropriately in context. Although a model may be able to describe the meaning of a proverb, this ability does not necessarily indicate that it can identify an appropriate proverb for a particular situation. Research on culturally grounded figurative language has similarly examined the relationship between figurative meaning interpretation and context-sensitive language use (Attia et al., 2026; Guo et al., 2025). Therefore, evaluating proverb understanding only through explanation generation may provide an incomplete picture of a model’s semantic understanding of proverbs.

Moreover, it remains unclear how LLM performance differs between interpretation and context application when processing Taiwanese proverbs. Investigating this issue may provide a more detailed understanding of the capabilities and limitations of LLMs in processing culturally embedded figurative language.

To address these gaps, this thesis adopts a framework involving two evaluation tasks for Taiwanese proverbs. Specifically, it pairs an open-ended Interpretation task, which evaluates the generation of proverb explanations using BERTScore-F1 (Zhang et al., 2020), with a Context Application task involving four answer choices, which evaluates whether a model can select an appropriate proverb for a given situation using accuracy. By examining both explanation generation and context-sensitive proverb selection, this framework provides a more comprehensive assessment of LLMs’ interpretation and semantic understanding of culturally embedded Taiwanese proverbs.

1.3 Research Questions

This thesis investigates how effectively contemporary LLMs understand and apply Taiwanese proverbs in English contexts. The study addresses the following main research question and sub-questions:

Main Research Question: How do contemporary LLMs perform across different evaluation tasks for interpreting and applying Taiwanese proverbs?

RQ1. Interpretation: How accurately can LLMs interpret Taiwanese proverbs?

RQ2. Context Application: How accurately can LLMs select and apply Taiwanese proverbs in appropriate contexts?

RQ3. Task Divergence and Model Comparison: How do LLM performances and relative rankings differ between the Interpretation task and the Context Application task?

1.4 Contributions and Thesis Structure

This thesis makes three main contributions. First, it provides a systematic evaluation of LLMs on a dataset of 468 Taiwanese proverbs, focusing on their ability to interpret proverb meanings in English.

Second, it extends proverb evaluation beyond interpretation by introducing a Context Application task based on 100 context-based questions.

Third, it compares four LLMs across these two complementary tasks, providing a more detailed assessment of their ability to understand and appropriately apply Taiwanese proverbs.

The remainder of this thesis is organized into five chapters. Chapter 2 presents related work of figurative language and LLM evaluation. It also introduces BERTScore, which is used to evaluate the generated proverb explanations.

Chapter 3 describes the methodology of the study, including data collection, dataset construction, the two evaluation tasks, the LLMs evaluated, the experimental setup, the evaluation methods, and the statistical analyses.

Chapter 4 presents the results of the experiments, including the descriptive and statistical results for the Interpretation task and the Context Application task. It also compares the performance of open-source and closed-source LLMs and summarizes the overall findings.

Chapter 5 discusses the results in detail, focusing on model performance in the two evaluation tasks, task-dependent differences in performance, and comparisons between open-source and closed-source models. It also examines the linguistic and cultural challenges involved in evaluating Taiwanese proverbs and discusses the implications of the findings for LLM evaluation.

Finally, Chapter 6 concludes the thesis by summarizing the main findings and answering the research questions. It also presents the contributions of the study, discusses its limitations, and suggests directions for future research.

Chapter 2

Related Work

This chapter reviews the theoretical background and previous research relevant to evaluating LLMs on culturally embedded figurative language, with a particular focus on Taiwanese proverbs. It introduces key concepts in figurative language and discusses the linguistic and cultural characteristics of proverbs, including Taiwanese proverbs. It then examines previous research on how LLMs understand figurative language and discusses approaches to evaluating LLM performance. Finally, it introduces BERTScore, the metric used in this study to evaluate model-generated proverb explanations.

2.1 Figurative Language

Figurative language is a foundational component of human communication that includes stylistic devices such as metaphors, idioms, irony, hyperbole, and proverbs (Colston, 2015). Unlike literal expressions, whose meanings are generally interpreted through the combination of their constituent meanings, figurative expressions may involve additional interpretive processes that extend beyond their literal meanings (Vulchanova et al., 2019).

In NLP, understanding figurative language has long been a challenge for computational systems (Martin, 1996). Traditional computational approaches to figurative language have often focused on linguistic and conceptual knowledge, but interpreting figurative expressions remains challenging when their intended meanings depend on contextual expectations and knowledge beyond their literal word meanings (Loenneker-Rodman & Narayanan, 2012; Martin, 1996). This issue is particularly relevant to proverbs, whose interpretation may require knowledge of their conventional meanings as well as the cul-

tural and pragmatic contexts in which they are used (Mieder, 2004). The interpretation of proverbs may involve metaphor, irony, historical context, and shared cultural knowledge, requiring readers or listeners to move beyond surface-level lexical semantics (Norrick, 1985). As a result, proverbs provide a valuable domain for investigating how computational systems process figurative language and culturally situated meanings.

Taiwanese Taigi includes culturally specific words and expressions documented in the Dictionary of Frequently-Used Taiwanese Taigi (Ministry of Education, 2011b). More generally, research on proverbs has shown that their interpretation and use can be influenced by contextual and cultural factors (H.-l. Lai & Hsu, 2025). This makes Taiwanese Taigi proverbs a useful domain for examining how computational models interpret figurative language and capture culturally grounded meanings.

With the rapid advancement of LLMs, NLP systems have demonstrated significant improvements in a wide range of tasks (Ahuja et al., 2024). However, strong performance on general language understanding benchmarks does not necessarily indicate robust competence in interpreting figurative language. Recent research suggests that LLM performance on metaphor interpretation may be influenced by surface-level features, such as lexical overlap and sentence length, rather than by metaphorical content alone (Sanchez-Bayona & Agerri, 2025). Moreover, evaluating culturally grounded figurative language requires attention to both semantic interpretation and pragmatic use, as models may perform differently when explaining an expression and applying it appropriately in context (Attia et al., 2026; Guo et al., 2025). This issue is particularly important for proverbs and figurative expressions from culturally specific or low-resource linguistic backgrounds, where successful interpretation may depend on knowledge that is not adequately captured by general language understanding benchmarks (Al Sefat, 2025; Wang et al., 2025).

2.2 LLM Evaluation

LLM evaluation has expanded beyond traditional performance metrics on static benchmarks to encompass reasoning, instruction following, and domain-specific capabilities (Zhao et al., 2026). Evaluation tasks can generally be divided into closed-ended and open-ended formats. Closed-ended tasks, such as multiple-choice question answering, allow for relatively objective and consistent scoring, whereas open-ended tasks assess a

model’s ability to generate explanations and articulate concepts.

For culturally embedded language, relying on a single task format may provide an incomplete picture of model capabilities. A model may correctly identify the meaning of a proverb in a multiple-choice task but fail to explain its cultural implications or apply it appropriately in context. Therefore, evaluating both the interpretation and contextual application of figurative expressions can provide a more comprehensive assessment of LLM performance (Guo et al., 2025).

2.3 BERTScore for Evaluating Generated Explanations

Automatic evaluation of open-ended text generation presents several methodological challenges, particularly when generated responses may differ from reference texts in wording while conveying similar meanings. This issue is especially relevant to proverb interpretation, as multiple English explanations of the same Taiwanese proverb may express the intended meaning using different words or sentence structures.

BERTScore is an embedding-based evaluation metric that measures semantic similarity between candidate and reference texts using contextual embeddings (Zhang et al., 2020). It computes precision, recall, and F1 scores based on token-level similarities between the contextualized representations of the candidate and reference texts. By considering contextual semantic representations rather than relying only on exact lexical matching, BERTScore can accommodate variations in wording between generated and reference texts.

In this study, BERTScore-F1 is used to evaluate the explanations generated in the Interpretation task. It measures how similar the meanings of model-generated explanations are to those of reference explanations, even when they use different words or sentence structures. However, BERTScore measures textual semantic similarity and does not necessarily capture all aspects of proverb interpretation, such as cultural knowledge, pragmatic appropriateness, or the completeness of an explanation. These limitations are taken into account when interpreting the evaluation results.

Chapter 3

Methodology

3.1 Data Collection

The *Dictionary of Frequently-Used Taiwanese Taigi* was purposively selected as the primary data source because it is the first government-compiled Taiwanese Taigi dictionary and contains a collection of commonly used Taiwanese Taigi words and expressions (Ministry of Education, 2011a). The term *Taiwanese Taigi* is retained here as part of the official English name of the dictionary (Ministry of Education, 2011a). The dictionary provides information on Taiwanese Taigi words and expressions, including characters, pronunciation, Mandarin translations, synonyms, antonyms, and example sentences to facilitate learning and comprehension (Ministry of Education, 2011b).

The proverb data were obtained from the proverb category of the dictionary (Ministry of Education, n.d.). All proverb entries listed in this category were extracted, resulting in a total of 468 Taiwanese proverbs. Since the study aimed to evaluate the complete set of proverbs available in the dictionary’s proverb category, no additional selection or filtering criteria were applied.

The extracted proverbs were organized according to the alphabetical order of the first letter of their Taiwanese Romanization. This ordering was retained to provide a consistent structure for the dataset and to facilitate subsequent data processing and analysis.

No duplicate removal was necessary because no duplicate proverb entries were identified in the dictionary or in the data collected through web scraping. All 468 entries were therefore retained in the final dataset. This collection of 468 proverbs constitutes the full dataset used in this study.

For the Context Application task, evaluating all 468 proverbs would have resulted in a substantially larger evaluation set. Therefore, a simple random sample of 100 proverbs was drawn from the full dataset of 468 Taiwanese proverbs. Sampling was performed without replacement using a fixed random seed (`random_state = 42`) to ensure reproducibility. The resulting sample of 100 proverbs was used to construct the context-based questions described in the following sections.

3.2 Dataset Description

Two datasets were used for the evaluation conducted in this study. The first dataset, referred to as the Interpretation Dataset, contains all 468 Taiwanese proverbs collected from the proverb category of the *Dictionary of Frequently-Used Taiwanese Taigi*, together with their corresponding English reference interpretations. This dataset is used to evaluate proverb interpretation. The second dataset, referred to as the Context Application Dataset, consists of 100 proverbs randomly sampled from the full set of 468 proverbs. These proverbs were used to construct 100 multiple-choice questions based on conversational contexts for the Context Application task.

The two datasets were designed to examine complementary aspects of proverb understanding. The Interpretation Dataset focuses on whether a model can identify and express the intended meaning of an individual proverb, while the Context Application Dataset examines whether a model can identify an appropriate proverb for a given conversational situation.

3.2.1 Interpretation Dataset

The `id` field provides a unique identifier for each proverb. The `proverb` field contains the original proverb as provided by the dictionary. The `definition_zh` field contains the Chinese definition of the proverb. The `interpretation_ref` field contains the English reference interpretations used as the reference text for the automatic evaluation of model-generated interpretations. Finally, the `url` field records the corresponding dictionary webpage for each proverb.

Table 3.1 presents an example of the structure of the Interpretation Dataset.

Table 3.1: Examples of the Interpretation Dataset

id	proverb	definition_zh	interpretation_ref	url
1	鴨仔聽雷。	指鴨子聽到雷聲，並不知道是怎麼回事。比喻一個人對所接收的訊息無法理解。	It refers to someone who hears or receives information but is unable to understand its meaning.	Dictionary entry
2	愛拚才會贏。	要努力才會贏。比喻唯有努力奮鬥，才能克服困難，成就一番事業。	Only through hard work and perseverance can people overcome difficulties and achieve success.	Dictionary entry
3	愛嬌毋驚流鼻水。	為了展示自己的好看的身材，天氣再冷，寧可挨寒受凍，也不願多穿衣服。	To show off an attractive figure, a person would rather endure the cold than wear more clothing.	Dictionary entry

3.2.2 Context Application Dataset

The Context Application Dataset consists of 100 multiple-choice questions designed to evaluate whether LLMs can select an appropriate Taiwanese proverb for a given conversational context. Unlike the Interpretation tasks that assess direct definition recall, this task tests a model’s pragmatic ability to apply proverb understanding within authentic everyday scenarios.

Each entry contains a context identifier (`context_id`), a proverb identifier (`proverb_id`), the target proverb (`proverb`), a conversational context (`conversation`), four candidate options (`option_A` through `option_D`), and the correct answer (`correct_answer`).

To prevent models from relying on simple keyword matching or superficial elimination, the three distractors for each question were manually selected from the remaining pool of 467 proverbs. Candidate distractors were retrieved by searching for related keywords and topics across the dataset. These options were then manually reviewed to ensure they were semantically plausible within the dialogue yet strictly incorrect based on their precise meanings. Exact synonyms were excluded to guarantee a single unambiguous correct answer. To avoid positional bias, the position of the correct answer was assigned across options A through D at random via an online tool.¹

For example, as shown in Table 3.2, one question presents a conversation where a speaker warns against accusing someone without proof. While Option C (疑人，毋成賊) is the correct match, the distractors include proverbs related to spreading rumors

¹<https://www.gigacalculator.com/randomizers/random-letter-generator.php>

(Option A, 講一个影，生一个囡), frequently making verbal mistakes (Option B, 三講，四毋著), or misperceptions (Option D, 目睷花花，匏仔看做菜瓜). This design forces the model to evaluate subtle pragmatic differences rather than relying on broad thematic associations.

Table 3.2 presents an example entry from the Context Application Dataset.

Table 3.2: Example of the Context Application Dataset

Field	Example
context_id	CAP001
proverb_id	55
proverb	疑人，毋成賊。
conversation	Tom: “The manager wants to confront Lisa because she was the last person in the office before the money disappeared.” Sarah: “Let’s talk to everyone who was there first. We don’t know what happened yet.”
option_A	講一个影，生一个囡。
option_B	三講，四毋著。
option_C	疑人，毋成賊。
option_D	目睷花花，匏仔看做菜瓜。
correct_answer	C

3.2.3 Interpretation References

To evaluate the English interpretations generated by the LLMs, a reference interpretation was established for each of the 468 Taiwanese proverbs in the dataset. Proverbs represent a form of figurative language that is deeply culturally embedded, meaning their intended meanings cannot always be inferred only from literal wording (Mieder, 2004). Consequently, the reference texts were designed to capture the intended pragmatic meaning of each proverb rather than providing a literal translation (Norrick, 1985).

The reference creation process followed a systematic pipeline based on the original Chinese dictionary definitions (`definition_zh`). First, initial English drafts were produced from the Chinese definitions using DeepL Translator.² Next, these initial texts underwent thorough manual review and manual editing to correct inaccuracies, eliminate translation artifacts, and ensure that each entry accurately conveyed the intended figurative meaning.

Finally, the manually revised English texts were recorded in the `interpretation_ref`

²<https://www.deepl.com/en/translator>

field of the dataset alongside their corresponding proverb identifiers. During the evaluation phase, these finalized references served as the benchmark texts against which model outputs were evaluated using BERTScore-F1. The interpretations produced by each LLM were subsequently collected in the **Interpretation Model Evaluation Dataset**.

3.3 Models and Experimental Setup

Four LLMs were evaluated in this study: Qwen 3.7, GPT-5.6 Luna, Gemini 3.6 Flash, and DeepSeek. These four models were selected to represent the open-source and closed-source categories examined in this study. All evaluations were conducted directly through the official web interfaces of the respective models. Table 3.3 summarizes the models, platforms, and access dates.

Table 3.3: LLMs Used in the Study

Model	Platform	Access Date
Qwen 3.7	Alibaba	July 28–September 8
GPT-5.6 Luna	OpenAI	July 28–September 8
Gemini 3.6 Flash	Google	July 28–September 8
DeepSeek	DeepSeek	July 28–September 8

The same experimental setup was applied to all four models to ensure comparability. For the Interpretation task, each model was evaluated on all 468 Taiwanese proverbs in the dataset. The same prompt was used for all models. The prompt asked the models to explain the figurative meaning of each proverb in one short sentence in English. The models were instructed to focus on the core figurative meaning, avoid literal interpretation and word-for-word translation, and provide only the explanation.

For the Context Application task, each model was evaluated on the same set of 100 questions. The same prompt was used for all models. Each question presented a situation and four Taiwanese proverb options. The models were asked to select the proverb that best matched the situation and to consider the intended meaning of the proverbs rather than relying only on literal word overlap. The models were instructed to output only the letter of the selected answer, namely A, B, C, or D.

To minimize post-processing artifacts and ensure output compliance in the Context Application task, the prompt explicitly instructed models to output only the letter of the selected option. Under these constrained instructions, all four models achieved complete

compliance across all 100 test items, generating single-letter responses without explanatory text or formatting errors. Nevertheless, a standardized post-processing protocol was established prior to evaluation to handle potential edge cases consistently. Under this protocol, any extra explanatory text following a valid option letter would be disregarded, retaining only the initial candidate letter. Furthermore, any missing or unparseable output would be treated as an invalid response and scored as incorrect without re-prompting or retrying.

Across both tasks, each model was run once for each test item under identical conditions, and the resulting raw outputs were saved for evaluation. The complete prompts used for both tasks are provided in Appendix A.

3.4 Evaluation Methods

Two distinct evaluation methods were used to assess the performance of the LLMs across the two tasks. The Interpretation task was evaluated using BERTScore (Zhang et al., 2020), while the Context Application task was evaluated using accuracy.

For the Interpretation task, BERTScore was selected because the aim was to assess whether the English explanations generated by the models conveyed meanings similar to those of the reference interpretations, rather than whether they reproduced the same wording. Unlike metrics based primarily on lexical or n-gram overlap, BERTScore uses contextual embeddings to measure the semantic similarity between generated and reference texts (Zhang et al., 2020). This makes it suitable for evaluating proverb interpretations, as the same figurative meaning may be expressed using different vocabulary or sentence structures. However, BERTScore measures the similarity between a generated interpretation and a reference interpretation, and therefore does not directly capture all aspects of proverb understanding, such as cultural appropriateness, pragmatic meaning, or completeness.

Each English interpretation generated by a model was evaluated against its corresponding reference interpretation. For each proverb and model pairing, BERTScore precision, recall, and F1 scores were calculated. BERTScore-F1 was used as the primary metric for comparing model performance on the Interpretation task.

For the Context Application task, each model evaluated the same set of 100 multiple-

choice questions. A response was considered correct when the model selected the predefined correct proverb option. Model performance was quantified using accuracy, defined as the percentage of correctly answered questions out of the 100 total items.

3.5 Statistical Analysis

Statistical analyses were conducted separately for the proverb interpretation and Context Application tasks because the two tasks produce different types of evaluation data.

For the Interpretation task, continuous BERTScore-F1 scores were collected across all 468 proverbs for each of the four LLMs. A Friedman test was used to determine whether there were statistically significant differences in BERTScore-F1 among the four models. The Friedman test was selected because the same set of proverbs was evaluated by all four models, resulting in related observations across models. When the overall test was significant, pairwise comparisons were conducted to identify differences between individual models. The Holm correction was applied to adjust for multiple comparisons.

For the Context Application task, each of the 100 questions produced a binary outcome indicating whether the model selected the correct proverb. Because the same questions were answered by all four models, a Cochran’s Q test was used to examine whether the models differed significantly in their overall performance. When the overall test was significant, pairwise McNemar tests were conducted to compare individual models. The Holm correction was applied to the resulting pairwise comparisons to control for multiple testing.

A significance level of $\alpha = 0.05$ was used for all statistical tests.

Chapter 4

Results

4.1 Interpretation Results

The Interpretation task evaluated the ability of the four LLMs to generate English interpretations of 468 Taiwanese proverbs. The interpretations generated by the models were compared with the corresponding English reference interpretations using BERTScore. BERTScore precision, recall, and F1 scores were calculated for each proverb and each model, with BERTScore-F1 used as the primary measure for comparing model performance.

4.1.1 Descriptive Results

Table 4.1 presents the descriptive statistics for BERTScore precision, recall, and F1 scores for the four LLMs. BERTScore-F1 was used as the primary measure because it combines semantic precision and recall into a single score.

Table 4.1: Descriptive Statistics of BERTScore for the Interpretation task

Model	N	Precision	Recall	F1	SD of F1
Qwen 3.7	468	0.9010	0.9016	0.9012	0.0200
GPT-5.6 Luna	468	0.8971	0.8925	0.8947	0.0240
Gemini 3.6 Flash	468	0.8877	0.8842	0.8859	0.0206
DeepSeek	468	0.8913	0.8741	0.8825	0.0215

Qwen 3.7 achieved the highest mean BERTScore-F1 (0.9012), followed by GPT-5.6 Luna (0.8947), Gemini 3.6 Flash (0.8859), and DeepSeek (0.8825). The difference between the highest and lowest mean BERTScore-F1 scores was 0.0187, indicating that

the overall differences in semantic similarity between the models were relatively small in absolute terms. The standard deviations of BERTScore-F1 ranged from 0.0200 to 0.0240, suggesting that the variation in scores across the 468 proverbs was similar among the four models.

Qwen 3.7 obtained the highest mean BERTScore precision (0.9010) and recall (0.9016), indicating that it achieved the highest scores on both components of the metric. GPT-5.6 Luna ranked second in both precision (0.8971) and recall (0.8925). However, the ranking of the remaining two models differed between the two measures. DeepSeek obtained a higher mean precision score (0.8913) than Gemini 3.6 Flash (0.8877), whereas Gemini 3.6 Flash achieved a higher mean recall score (0.8842) than DeepSeek (0.8741).

The difference between precision and recall was also different across the models. Qwen 3.7 showed nearly identical precision and recall scores, with a difference of 0.0006. GPT-5.6 Luna had a precision score that was 0.0046 higher than its recall score. Gemini 3.6 Flash showed a difference of 0.0035 between precision and recall, while DeepSeek showed the largest difference of 0.0172. This indicates that DeepSeek’s precision and recall scores were less balanced than those of the other three models.

4.1.2 Statistical Results

A Friedman test was conducted to determine whether the four LLMs differed significantly in their BERTScore-F1 performance across the 468 Taiwanese proverbs. The test revealed a statistically significant difference among the models ($\chi^2(3) = 292.43$, $p < .001$; see Table 4.2). This result indicates that at least one model differed significantly from the others, but it does not identify which specific pairs of models differed.

Table 4.2: Friedman Test Results for the Interpretation task

Test	N	Number of Models	χ^2	df	p
Friedman test	468	4	292.43	3	< .001

Because the overall Friedman test was significant, pairwise Wilcoxon signed-rank tests were conducted to identify differences between individual models. The Holm correction was applied to control for multiple comparisons. All six pairwise comparisons were statistically significant after correction (see Table 4.3).

Table 4.3: Pairwise Wilcoxon Signed-Rank Tests for the Interpretation task

Comparison	Statistic	Holm-adjusted p	Significant
GPT-5.6 Luna vs. Gemini 3.6 Flash	28545	$< .001$	Yes
GPT-5.6 Luna vs. Qwen 3.7	41626	.000012	Yes
GPT-5.6 Luna vs. DeepSeek	21280	$< .001$	Yes
Gemini 3.6 Flash vs. Qwen 3.7	15811	$< .001$	Yes
Gemini 3.6 Flash vs. DeepSeek	44404	.000452	Yes
Qwen 3.7 vs. DeepSeek	14526	$< .001$	Yes

These results indicate that the four LLMs differed significantly from one another in their BERTScore-F1 performance across the 468 Taiwanese proverbs.

4.2 Context Application Results

The Context Application task evaluated whether the four LLMs could select the Taiwanese proverb that best matched a given conversational situation. The evaluation consisted of 100 questions, with each question presenting four proverb options and one pre-determined correct answer. Model performance was measured using accuracy, defined as the percentage of correctly answered questions.

4.2.1 Descriptive Results

Table 4.4 presents the accuracy of the four LLMs on the Context Application task.

Table 4.4: Accuracy of the LLMs on the Context Application task

Model	N	Correct Answers	Accuracy (%)
Gemini 3.6 Flash	100	98	98
DeepSeek	100	94	94
Qwen 3.7	100	94	94
GPT-5.6 Luna	100	83	83

Gemini 3.6 Flash achieved the highest accuracy, correctly answering 98 of the 100 questions (98%). Qwen 3.7 and DeepSeek each answered 94 questions correctly (94%), while GPT-5.6 Luna correctly answered 83 questions (83%).

4.2.2 Statistical Results

A Cochran’s Q test was conducted to determine whether the four LLMs differed significantly in their performance on the 100 Context Application questions. The test revealed a statistically significant difference among the models, $Q(3) = 27.22$, $p < .001$.

Because the overall Cochran’s Q test was significant, pairwise McNemar tests were conducted to identify differences between individual models. The Holm correction was applied to control for multiple comparisons (see Table 4.5).

Table 4.5: Pairwise McNemar Test Results for the Context Application task

Comparison	Statistic	Holm-adjusted p	Significant
GPT-5.6 Luna vs. Gemini 3.6 Flash	0	.000366	Yes
GPT-5.6 Luna vs. Qwen 3.7	2	.029541	Yes
GPT-5.6 Luna vs. DeepSeek	0	.004883	Yes
Gemini 3.6 Flash vs. Qwen 3.7	0	.375	No
Gemini 3.6 Flash vs. DeepSeek	0	.375	No
Qwen 3.7 vs. DeepSeek	3	1.000	No

The results showed statistically significant differences between GPT-5.6 Luna and Gemini 3.6 Flash ($p_{\text{Holm}} = .000366$), GPT-5.6 Luna and Qwen 3.7 ($p_{\text{Holm}} = .029541$), and GPT-5.6 Luna and DeepSeek ($p_{\text{Holm}} = .004883$).

Taken together, these results indicate that the performance of GPT-5.6 Luna differed significantly from that of each of the other three models, whereas the differences among Gemini 3.6 Flash, Qwen 3.7, and DeepSeek were not statistically significant. Consistent with the descriptive results, GPT-5.6 Luna had a lower accuracy than each of the other three models.

4.3 Comparison Between open-source and closed-source LLMs

The four LLMs were also compared according to their source availability. Qwen 3.7 and DeepSeek were classified as open-source models, whereas GPT-5.6 Luna and Gemini 3.6 Flash were classified as closed-source models. The comparison was conducted across both the Interpretation and Context Application tasks. Because each group contained only two models, this comparison serves as a descriptive analysis of the evaluated models rather than a generalizable inferential comparison between model architectures.

Table 4.6 summarizes the mean performance metrics for both model families across the two evaluation tasks.

Table 4.6: Descriptive Comparison Between Open-Source and Closed-Source Models

Task	Open-Source Mean	Closed-Source Mean	Difference
Interpretation (BERTScore-F1)	0.8919	0.8903	0.0016
Context Application (Accuracy)	94.0%	90.5%	3.5%

In the Interpretation task, the open-source and closed-source model groups demonstrated highly comparable semantic alignment, with mean BERTScore-F1 scores of 0.8919 and 0.8903, respectively. The marginal difference of 0.0016 indicates that both model families perform almost identically when generating free-form figurative interpretations.

In contrast, a more visible performance disparity emerged in the Context Application task. The open-source models achieved a mean accuracy of 94.0%, outperforming the closed-source models (90.5%) by 3.5 percentage points. Overall, the performance gap between open-source and closed-source models was noticeably more pronounced in pragmatic contextual application than in open-ended figurative interpretation.

Chapter 5

Discussion

5.1 Interpretation Performance

The Interpretation task results show that Qwen 3.7 achieved the highest mean BERTScore-F1 among the four LLMs, followed by GPT-5.6 Luna, Gemini 3.6 Flash, and DeepSeek. The differences between all model pairs were statistically significant after Holm correction, confirming that the performance ranking observed in the descriptive statistics was robust across pairwise comparisons.

The relatively small magnitude of difference in mean BERTScore-F1 is equally important when interpreting these findings. Although the performance gaps were statistically significant, all four models achieved mean BERTScore-F1 values above 0.88 (see Table 4.1). This indicates that all models consistently generated interpretations that were highly semantically similar to the English reference interpretations. The results therefore show a consistent performance difference among the models rather than a large separation in overall interpretation quality.

However, BERTScore-F1 should not be interpreted as a complete measure of proverb comprehension. The metric evaluates the semantic similarity between a model-generated interpretation and a reference interpretation. A high score therefore indicates similarity to the meaning represented in the reference, but does not necessarily demonstrate that the model has captured every cultural, pragmatic, or contextual aspect of a proverb.

These findings suggest that Qwen 3.7 was the most effective model under the reference-based evaluation framework employed in this study. At the same time, the results should be contextualized within the specific characteristics of the English reference interpreta-

tions and the automated evaluation pipeline.

5.2 Context Application Performance

The Context Application task results show a different performance pattern from the Interpretation task. Gemini 3.6 Flash achieved the highest accuracy (98%), whereas GPT-5.6 Luna obtained the lowest (83%; see Table 4.4). Cochran’s Q test showed a statistically significant difference among the four models, and the pairwise comparisons showed that GPT-5.6 Luna differed significantly from each of the other three models.

The high accuracy achieved by Gemini 3.6 Flash suggests that the model was effective at selecting a proverb that matched the given conversational situations. The task required the models to distinguish between four candidate proverbs and identify the option whose intended meaning best matched the situation. This differs from the Interpretation task, where the models generated an English explanation of an individual proverb.

Qwen 3.7 and DeepSeek showed identical performance on this task, with both models achieving an accuracy of 94%, and their difference was not statistically significant. In contrast, GPT-5.6 Luna performed significantly worse than all competing models.

The Context Application results should nevertheless be interpreted in relation to the task design. Each question provided four predefined proverb options and one correct answer. The results therefore demonstrate the models’ ability to distinguish an appropriate proverb from the alternatives provided in the dataset. They do not necessarily demonstrate how the models would select or generate a proverb in an unrestricted conversational setting.

5.3 Task-Dependent Model Performance

A central finding of this study is that the relative performance of the models varied across the two evaluation tasks. Qwen 3.7 achieved the highest performance on the Interpretation task, whereas Gemini 3.6 Flash achieved the highest performance on the Context Application task. The model rankings were therefore not consistent across the two tasks.

The difference is particularly visible in the performance of GPT-5.6 Luna. The model

ranked second on the Interpretation task but achieved the lowest accuracy on the Context Application task. Qwen 3.7 also provides an important example: it achieved the highest BERTScore-F1 on the Interpretation task but did not achieve the highest accuracy on the Context Application task. These patterns indicate that performance on one evaluation task does not necessarily predict performance on another.

This difference can be related to the distinct requirements of the two tasks. The Interpretation task required the models to express the meaning of a proverb in English, whereas the Context Application task required them to identify an appropriate proverb for a particular situation. The latter task involved distinguishing among competing candidate meanings within a given context. These requirements represent related but distinct aspects of proverb understanding.

The findings therefore support the use of multiple complementary tasks when evaluating LLM understanding of culturally embedded figurative language. Treating proverb understanding as a single capability may obscure meaningful differences in model performance across different types of evaluation.

5.4 open-source and closed-source Models

The comparison between open-source and closed-source models showed broadly similar performance on the Interpretation task. The open-source models achieved a mean BERTScore-F1 of 0.8919, compared with 0.8903 for the closed-source models (see Table 4.6). The difference was therefore very small.

A somewhat larger difference was observed on the Context Application task. The open-source models achieved a mean accuracy of 94.0%, compared with 90.5% for the closed-source models (see Table 4.6). However, this difference should not be interpreted as evidence that open-source models generally perform better than closed-source models.

The main reason for this limitation is the small number of models in each group. Only two open-source and two closed-source models were included in the study. The observed group differences may therefore reflect characteristics of the individual models rather than a general difference between the two categories. Factors such as model architecture, training data, and model size are also not isolated in this comparison.

The open-source and closed-source results should therefore be treated as descriptive

observations within the present model set. A broader comparison including more models from both categories would be necessary to determine whether similar performance patterns occur more generally.

5.5 Linguistic and Cultural Challenges

Taiwanese proverbs provide a challenging setting for LLM evaluation because their intended meanings are not always directly recoverable from their literal wording. Understanding such expressions may require the model to connect the linguistic form of the proverb with knowledge about its intended meaning and use.

Cultural knowledge is particularly relevant because proverbs can reflect experiences, values, and social practices associated with the communities in which they are used. The Context Application task further highlights the importance of this aspect because the model had to determine which proverb was appropriate for a particular conversational situation rather than simply explain an isolated expression.

The use of English as the output language introduces an additional linguistic challenge. The original expressions are Taiwanese Taigi, while the interpretations and reference texts used in the Interpretation task are in English. The models therefore had to identify the intended meaning of the original expression and express that meaning in another language. This may introduce semantic shifts between the meaning of the original expression and its English representation.

These characteristics suggest that culturally embedded proverbs can test several aspects of language understanding at the same time. However, the present study does not isolate individual linguistic or cultural factors. Therefore, the results cannot be used to determine which specific linguistic or cultural characteristics were responsible for differences in model performance.

5.6 Implications for LLM Evaluation

The findings have several implications for the evaluation of LLMs on culturally embedded figurative language. First, the different model rankings across the two tasks suggest that evaluating only one aspect of proverb understanding may provide an incomplete

picture of model performance. A model that performs well when generating a semantic interpretation may not perform equally well when selecting a proverb for a particular situation.

Second, the results support the use of complementary evaluation tasks. The Interpretation task examined whether models could express the intended meaning of individual proverbs, while the Context Application task examined whether models could identify an appropriate proverb in context. Evaluating both tasks therefore provides a broader view of proverb understanding than either task alone.

Third, evaluation measures should be selected according to the requirements of the task. BERTScore-F1 was used to evaluate semantic similarity between generated interpretations and English reference interpretations, whereas accuracy was used for the selection-based Context Application task. Because these measures capture different properties of model performance, they should not be combined into a single overall score without a clear theoretical justification.

Finally, the findings suggest that culturally specific data can provide useful additional information for LLM evaluation. Taiwanese proverbs involve cultural and contextual knowledge, which may not be fully captured by general language benchmarks. Including such data in evaluation may therefore help examine how well models understand culturally embedded figurative language.

Chapter 6

Conclusion

6.1 Summary of Findings and Answers to Research Questions

This thesis investigated how contemporary LLMs understand Taiwanese proverbs across two complementary evaluation tasks: interpretation and contextual application. By evaluating four representative systems (Qwen 3.7, GPT-5.6 Luna, Gemini 3.6 Flash, and DeepSeek) on a dataset of 468 Taiwanese proverbs, this study yields the following principal findings.

Regarding the **Main Research Question**—*How do contemporary LLMs perform across different evaluation tasks for interpreting and applying Taiwanese proverbs?*—the four evaluated LLMs were assessed using two complementary tasks: Interpretation and Context Application. In the Interpretation task, the models achieved mean BERTScore-F1 scores ranging from 0.8825 to 0.9012, reflecting the semantic similarity between the generated explanations and the reference explanations according to BERTScore. In the Context Application task, accuracy ranged from 83% to 98%. Gemini 3.6 Flash achieved the highest accuracy at 98%, followed by Qwen 3.7 and DeepSeek at 94%, while GPT-5.6 Luna achieved 83%.

These results show that the relative performance of the models varied across the two evaluation tasks. In particular, the model with the highest accuracy in the Context Application task was not the model with the highest mean BERTScore-F1 in the Interpretation task. This finding underscores the importance of using complementary evaluation tasks

when examining LLMs’ interpretation and application of Taiwanese proverbs. Specifically, the detailed answers to the sub-questions are as follows:

First, regarding **RQ1 (Interpretation)**—*How accurately can LLMs interpret Taiwanese proverbs?*—all evaluated models achieved mean BERTScore-F1 scores ranging from 0.8825 to 0.9012, indicating that they were generally able to capture the intended figurative meaning of Taiwanese proverbs in English. Although statistically significant pairwise differences were observed, the tight performance cluster across systems suggests that the evaluated LLMs consistently captured the general intended meanings of the Taiwanese proverbs in the English translations.

Second, regarding **RQ2 (Context Application)**—*How accurately can LLMs select and apply Taiwanese proverbs in appropriate contexts?*—the evaluated models demonstrated generally high accuracy in selecting appropriate Taiwanese proverbs for given conversational contexts, with scores ranging from 83% to 98%. Gemini 3.6 Flash achieved the highest accuracy at 98%, followed by Qwen 3.7 and DeepSeek, both at 94%, while GPT-5.6 Luna achieved 83%. The post-hoc McNemar tests showed that GPT-5.6 Luna differed significantly from each of the other three models, while no significant differences were found among Gemini 3.6 Flash, Qwen 3.7, and DeepSeek. Compared with the Interpretation task, in which all pairwise model comparisons were statistically significant, the Context Application task revealed a more distinct performance pattern, with three models achieving similarly high accuracy and GPT-5.6 Luna showing a comparatively lower level of performance. This suggests that contextual application provides a different basis for distinguishing model performance from reference-based semantic interpretation.

Third, regarding **RQ3 (Task Divergence and Model Comparison)**—*How do LLM performances and relative rankings differ between the Interpretation task and the Context Application task?*—model performance patterns differed between the two evaluation tasks. Higher performance in the Interpretation task does not necessarily correspond to higher accuracy in the Context Application task. This suggests that interpreting the meaning of a proverb and selecting an appropriate proverb for a given context may involve related but distinct abilities. Furthermore, the descriptive comparison between open-source and closed-source models did not show a clear overall difference in performance across the two tasks.

6.2 Contributions

This thesis makes several key contributions to the empirical evaluation of LLMs on culturally embedded figurative language.

First, the study introduces an evaluation dataset consisting of 468 Taiwanese proverbs paired with English interpretations, along with a subset of 100 Taiwanese proverbs accompanied by context-based multiple-choice questions. By constructing the Context Application dataset, the study helps address an existing gap in figurative language benchmarks, providing a structured resource for evaluating how models handle culturally specific expressions that are often underrepresented in standard evaluation suites.

Second, the study establishes a dual task evaluation methodology that assesses proverb understanding through two complementary dimensions. The Interpretation task measures a model’s ability to express the intended meaning of an individual proverb in English, while the Context Application task measures its ability to identify an appropriate proverb for a given situation. This combined framework extends evaluation beyond simple interpretation matching toward contextual reasoning.

Third, the study demonstrates that model performance can vary substantially according to the evaluation tasks. Qwen 3.7 performed best on the Interpretation task, while Gemini 3.6 Flash performed best on the Context Application task. This finding highlights the importance of considering different aspects of language understanding when evaluating LLMs.

Fourth, the study offers practical contributions to low-resource language and culture preservation. By benchmark testing LLMs on Taiwanese proverbs, the findings highlight both the strengths and limitations of current models in preserving, translating, and accurately contextualizing low-resource cultural heritage in the English settings.

Finally, the study provides a comparison of open-source and closed-source models within the selected model set. Although this comparison is descriptive because of the small number of models, it provides an additional perspective on model performance across the two evaluation tasks.

6.3 Limitations

Several limitations should be considered when interpreting the findings and assessing the generalizability of this study.

First, only four LLMs, comprising two open-source and two closed-source models, were evaluated. Although these systems represent different model families and architectural approaches, the sample remains too small to generalize the findings to LLMs as a whole. The comparison between open-source and closed-source models should therefore be understood as a descriptive analysis of the selected systems rather than as evidence of general differences between the two categories.

Second, the Context Application task consisted of 100 questions selected from the full dataset of 468 Taiwanese proverbs. Although the questions were based on a random sample, the smaller sample limits the extent to which the results can represent the full proverb dataset. In addition, each question provided four predefined proverb options. Consequently, the task evaluated the selection of an appropriate proverb from a set of candidates rather than proverb generation in an unrestricted conversational setting.

Third, the conversational contexts and distractor options used in the Context Application task were constructed and reviewed by a single individual. Consequently, the intended uniqueness of the correct answer was not independently validated. This leaves open the possibility of ambiguity in the question design or alternative interpretations of the candidate options.

Fourth, the Interpretation task relied on English reference interpretations derived from Taiwanese dictionary definitions and subsequently reviewed and revised manually. Although this procedure provided a consistent basis for automatic evaluation, each reference represents only one possible English formulation of the intended meaning. Other valid English interpretations may therefore receive different BERTScore values.

Fifth, BERTScore evaluates semantic similarity between generated text and a reference text. It does not directly measure every aspect of proverb understanding, including cultural knowledge, cultural appropriateness, pragmatic meaning, or possible context-dependent interpretations. The interpretation results should therefore be understood as performance under the specific reference-based evaluation framework used in this study.

Sixth, computational efficiency was not systematically recorded across model runs. Therefore, the study does not provide a direct comparison of the computational efficiency

of the evaluated models.

Finally, each model was evaluated using a single generation pass per item in both tasks. The study therefore does not examine variation across repeated runs or different generation conditions. Differences in model outputs under other experimental settings may produce different performance estimates.

Taken together, these limitations mean that the findings should be interpreted as evidence of model performance under the specific datasets, tasks, references, and evaluation procedures used in this study. Broader conclusions would require more models, larger and validated evaluation datasets, repeated model runs, and additional evaluation methods.

6.4 Future Work

Building upon the empirical findings and methodological constraints of this study, future research can expand in several key directions. First, expanding the evaluation suite to include a broader collection of open-source and closed-source architectures will enable statistical comparison across model families.

Second, future studies could expand the Context Application dataset beyond the 100 questions evaluated in the present study. A larger sample would provide a more comprehensive assessment of contextual proverb understanding. Furthermore, future work could introduce open-ended contextual tasks in which models are required to generate an appropriate proverb rather than select one from a predefined set of candidate options.

Third, future research could enrich the evaluation of proverb interpretation by addressing the limitations noted in Section 6.3. Specifically, incorporating multiple independently produced English interpretations and human evaluation could capture dimensions beyond BERTScore, such as cultural appropriateness, pragmatic interpretation, and completeness of meaning. In addition, future work could systematically incorporate computational efficiency metrics, such as inference speed.

Fourth, future studies could examine the relationship between the linguistic and cultural characteristics of individual proverbs and model performance. Factors such as figurative structure, cultural specificity, lexical ambiguity, and contextual dependence could be analyzed to determine whether particular types of proverbs present greater challenges for LLMs.

Finally, evaluating output consistency through repeated runs across varying generation parameters, such as decoding temperatures, could provide deeper insights into model reliability and stability in minority language reasoning.

Overall, this thesis shows that LLMs were generally able to interpret and apply Taiwanese proverbs, while also demonstrating that performance depends on how proverb understanding is assessed. Future research that combines larger datasets, more models, human evaluation, and multiple evaluation tasks could provide a more complete understanding of the strengths and limitations of LLMs in culturally embedded language.

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References

- Ahuja, S., Aggarwal, D., Gumma, V., Watts, I., Sathe, A., Ochieng, M., Hada, R., Jain, P., Ahmed, M., Bali, K., & Sitaram, S. (2024). MEGEVERSE: Benchmarking large language models across languages, modalities, models and tasks. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 2598–2637). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.naacl-long.143>
- Al Sefat, A. (2025). BengaliFig: A low-resource challenge for figurative and culturally grounded reasoning in Bengali. In A. Shukla, S. Kumar, A. S. Bedi, & T. Chakraborty (Eds.), *Proceedings of the 1st workshop on multimodal models for low-resource contexts and social impact (mmloso 2025)* (pp. 58–77). Association for Computational Linguistics. <https://aclanthology.org/2025.mmloso-1.6/>
- Attia, M., Muhamed, A., Alkhamissi, M., Solorio, T., & Diab, M. T. (2026). Beyond understanding: Evaluating the pragmatic gap in LLMs’ cultural processing of figurative language. In V. Demberg, K. Inui, & L. Marquez (Eds.), *Proceedings of the 19th conference of the European chapter of the Association for Computational Linguistics (volume 1: Long papers)* (pp. 7238–7265). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2026.eacl-long.341>
- Colston, H. L. (2015). *Using figurative language*. Cambridge University Press. <https://doi.org/10.1017/CBO9781316226414>
- Guo, S., Zhuang, T., & Li, L. (2025). Proverbdualeval: A large-scale dataset for chinese proverb understanding and pragmatic ability evaluation. *Proceedings of the 2025 International Symposium on Artificial Intelligence and Computational Social Sciences*, 553–560. <https://doi.org/10.1145/3776759.3776846>

- Lai, H.-l., & Hsu, H.-L. (2025). Dog and cat proverbs. *Review of Cognitive Linguistics*. Published under the auspices of the Spanish Cognitive Linguistics Association, 23(2), 544–572. <https://doi.org/10.1075/rcl.00174.lai>
- Lai, Y.-C., Zheng, Y., Hsu, W.-H., Lin, Y., Cho, C.-H., Kuo, C.-C., Huang, C.-S., & Liao, Y.-F. (2024). Construction of large language models for taigi and hakka using transfer learning. *Proceedings of the 2024 International Conference on Oriental COCOSDA*. <https://doi.org/10.1109/O-COCOSDA64382.2024.10800470>
- Lien, C. (2003). In search of covert grammatical categories in taiwanese southern min: A cognitive approach to verb semantics. *Language and Linguistics*, 4(2), 379–402.
- Liu, C. C., Koto, F., Baldwin, T., & Gurevych, I. (2024). Are multilingual LLMs culturally-diverse reasoners? an investigation into multicultural proverbs and sayings. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 2016–2039). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.naacl-long.112>
- Loenneker-Rodman, B., & Narayanan, S. (2012). Computational approaches to figurative language. In *The cambridge handbook of psycholinguistics*. Cambridge University Press. <https://doi.org/10.1017/CBO9781139029377.025>
- Martin, J. H. (1996). Computational approaches to figurative language. *Metaphor and Symbolic Activity*, 11(1), 85–100. https://doi.org/10.1207/S15327868MS1101_5
- Mieder, W. (2004). *Proverbs: A handbook*. Greenwood Press.
- Ministry of Education. (2011a). Dictionary of frequently-used taiwanese taigi [Last accessed: September 13, 2026]. <https://sutian.moe.edu.tw/>
- Ministry of Education. (2011b). Dictionary of frequently-used taiwanese taigi: Introduction [Last accessed: September 13, 2026]. <https://sutian.moe.edu.tw/zh-hant/introduction/>
- Ministry of Education. (n.d.). Dictionary of frequently-used taiwanese taigi: Proverb category [Last accessed: September 13, 2026]. <https://sutian.moe.edu.tw/zh-hant/huliok/106/>
- Norricks, N. R. (1985). *How proverbs mean: Semantic studies in english proverbs*. Mouton Publishers.

- Sanchez-Bayona, E., & Agerri, R. (2025). Metaphor and large language models: When surface features matter more than deep understanding. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Findings of the association for computational linguistics: Acl 2025* (pp. 17462–17477). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-acl.898>
- Shutova, E. V. (2011, August). *Computational approaches to figurative language* (tech. rep. No. UCAM-CL-TR-803). University of Cambridge, Computer Laboratory. <https://doi.org/10.48456/tr-803>
- Vulchanova, M., Milburn, E., Vulchanov, V., & Baggio, G. (2019). Boon or burden? the role of compositional meaning in figurative language processing and acquisition. *Journal of Logic, Language and Information*, 28(2), 359–387. <https://doi.org/10.1007/s10849-019-09282-7>
- Wang, M., Pham, V. T., Moghimifar, F., & Vu, T.-T. (2025). Proverbs run in pairs: Evaluating proverb translation capability of large language model. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Findings of the association for computational linguistics: Acl 2025* (pp. 1646–1662). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.findings-acl.83>
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2020). Bertscore: Evaluating text generation with bert. *International Conference on Learning Representations (ICLR)*.
- Zhao, W. X., Zhou, K., Li, J., Tang, T., Dong, Z., Hou, Y., Zhang, B., Min, Y., Zhang, J., Liu, P., Wang, X., Du, Y., Yang, C., Chen, Y., Chen, Z., Jiang, J., Ren, R., Li, Y., Tang, X., ... Wen, J.-R. (2026). A survey of large language models. *Frontiers of Computer Science*, 20(12), 2012627. <https://doi.org/10.1007/s11704-026-60308-3>

Appendix A

Prompts

A.1 Proverb Interpretation Prompt

The following prompt was used for the Interpretation task:

You are given a Taiwanese proverb.

Proverb: "{PROVERB}"

Explain the figurative meaning of this proverb in one short sentence.

Requirements:

- Focus on the core figurative meaning.
- Do not interpret the proverb literally.
- Do not translate the proverb word by word.
- Avoid unnecessary elaboration beyond the core meaning.
- Keep the explanation concise.
- Write the explanation in English.
- Output only the explanation.

A.2 Context Application Prompt

The following prompt was used for the Context Application task:

You are given a situation and four Taiwanese proverbs.

Situation:

"{SCENARIO}"

Which proverb best fits this situation?

A. {OPTION_A}

B. {OPTION_B}

C. {OPTION_C}

D. {OPTION_D}

Requirements:

- Select the proverb that best matches the situation.
- Consider the intended meaning of each proverb.
- Do not choose an answer based only on literal word overlap.
- Output only the letter of the selected answer (A, B, C, or D).